

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 7: Ice and Rain Protection Systems

Syllabus Source: CAE Oxford (pp. 235-258) | Pass Target: >= 90% AviationExam

1. ANTI-ICING VS DE-ICING PHILOSOPHIES

Ice accumulation on aerodynamic surfaces reduces maximum lift by up to 30%, increases parasite drag by over 100%, and increases stall speed significantly. Commercial aircraft utilize two distinct operational methods:

- Anti-Icing (Preventative): Operates CONTINUOUSLY before entering suspected icing conditions (OAT $\leq +10^{\circ}\text{C}$ in visible moisture) to PREVENT any ice formation from ever adhering to protected surfaces.
- De-Icing (Remedial): Operates INTERMITTENTLY to break off and shed ice layers AFTER a predetermined thickness (e.g. 6 to 12 mm / 1/4 to 1/2 inch) has accumulated.

System Type	Operating Medium & Mechanism	Aircraft Protected Areas & Operational Limits
Thermal Anti-Ice	Hot engine bleed air ($\sim 200^{\circ}\text{C}$) piped through Piccolo tubes inside leading edge.	Slats and wing leading edges; engine nacelle intake cowlings. Engine anti-ice uses raw bleed air directly and operates independently of wing anti-ice.
Pneumatic De-Ice Boots	Inflatable neoprene rubber boots bonded to leading edges. Inflated cyclically by bleed air / vacuum pump.	Wings and tail empennage on turboprop and commuter aircraft. Must allow ice to build up before inflating; premature cycling can cause 'ice bridging' (ice forms a rigid shell around inflated boot).
Electrical Heating	Embedded resistive heating elements (AC or DC powered).	Pitot probes, static ports, TAT sensors, AOA vanes, propeller blade boots, flight deck windshields, and waste water drain masts.
Fluid (Weeping Wing)	Ethylene glycol fluid pumped through microscopic laser-drilled pores in titanium panels (TKS system).	General aviation and light turboprops. Provides both anti-icing and de-icing; limited by fluid reservoir quantity.

[!] EXAM TRAP (AVIATIONEXAM) - Engine Anti-Ice vs Wing Anti-Ice Selection Rules

- Engine Anti-Ice: MUST BE SELECTED ON whenever OAT (on ground) or TAT (in flight) is $+10^{\circ}\text{C}$ OR BELOW in visible moisture (clouds, fog with visibility $< 1,000$ m, rain, snow, slush)! - Wing Anti-Ice: Selected on in flight when ice accumulation is observed, or continuously in severe icing. Usually inhibited automatically on the ground via squat switch (except for brief functional test) to prevent overheating leading-edge slats!

2. WINDSCREEN HEATING & RAIN REMOVAL

Transport aircraft flight deck windshields are multi-laminated glass-vinyl sandwiches with two critical operational requirements:

- Electrical Demisting / De-icing: A transparent conductive metallic coating (e.g. gold film or Stannic oxide) laminated inside the glass is heated with AC power. Thermistors embedded in the vinyl modulate power to maintain $\sim 35^{\circ}\text{C}$ to 45°C . This prevents icing and demists the inner pane.

- Bird-Strike Impact Resistance: At cruising temperatures (-50°C), vinyl interlayers become brittle. Electrical heating softens the vinyl interlayer, making it pliable and ductile, allowing it to absorb a 4-lb bird impact at V_c without cockpit penetration!
- Rain Removal Systems: 1. Mechanical windshield wipers (limited to V_{max_wiper}, ~250 kt); 2. Rain repellent liquid sprayed onto glass (banned in dry conditions: smears into opaque grease); 3. Hydrophobic chemical surface coating that makes water bead off aerodynamically.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Anti-Ice vs De-Ice Systems

Rule 1 (Anti-Ice): PREVENTS ice from forming. Turned ON prior to entering icing conditions. Wing leading edges and engine nacelles use continuous hot engine bleed air. Rule 2 (De-Ice): REMOVES ice after accumulation. Pneumatic de-icer boots cycle periodically to break accumulated ice. If cycled too early, ice bridging may occur. Rule 3: Pitot probes, static ports, TAT probes, and windshields use electrical heating continuously in flight.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why must flight deck windshield electrical heating remain powered during normal cruise at high altitudes?

- [A]: To prevent cockpit glass fogging only
- [B]: To warm the vinyl interlayer, maintaining elasticity and shatterproof impact resistance against bird-strikes at freezing temperatures
- [C]: To supply emergency AC power
- [D]: To improve cockpit night vision

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At -50°C in cruise, cold vinyl is brittle. Windscreen heating keeps the vinyl core pliable and elastic, which is mandatory to certify bird-strike impact resistance under CS-25.

Q2: What operational hazard is associated with operating pneumatic de-ice boots prematurely before sufficient ice has accumulated?

- [A]: The rubber boot will blow off the wing
- [B]: Ice bridging: ice can freeze over the inflated boot shape, forming a hollow shell that subsequent boot cycles cannot break
- [C]: Engine bleed air pressure drops to zero
- [D]: The boots will overheat

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If inflated when only a thin slushy film exists, water can freeze over the extended contour, creating an 'ice bridge'. Subsequent inflations occur inside the cavity without shedding the outer ice shell.

Q3: When must engine cowl thermal anti-icing be turned ON in flight?

- [A]: Only when ice chunks are visibly flying into the engine
- [B]: Whenever Total Air Temperature (TAT) is +10°C or below and visible moisture (clouds, rain, snow) is present
- [C]: At all times above FL 250

- [D]: Only during descent

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under EASA standards, engine anti-ice must be on whenever TAT is +10°C or colder in any visible moisture, preventing ice buildup on cowl lips that could shed into engine fan blades.